

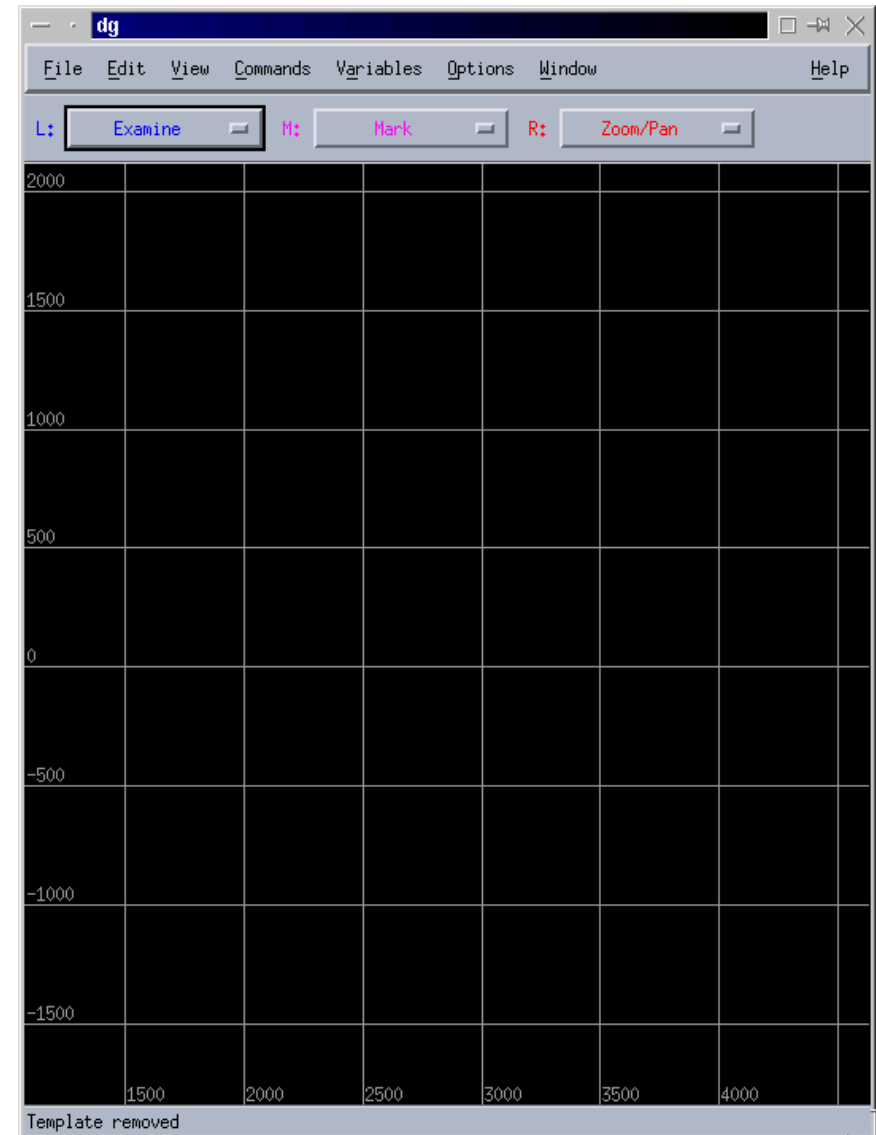
DG

David Coster, ...

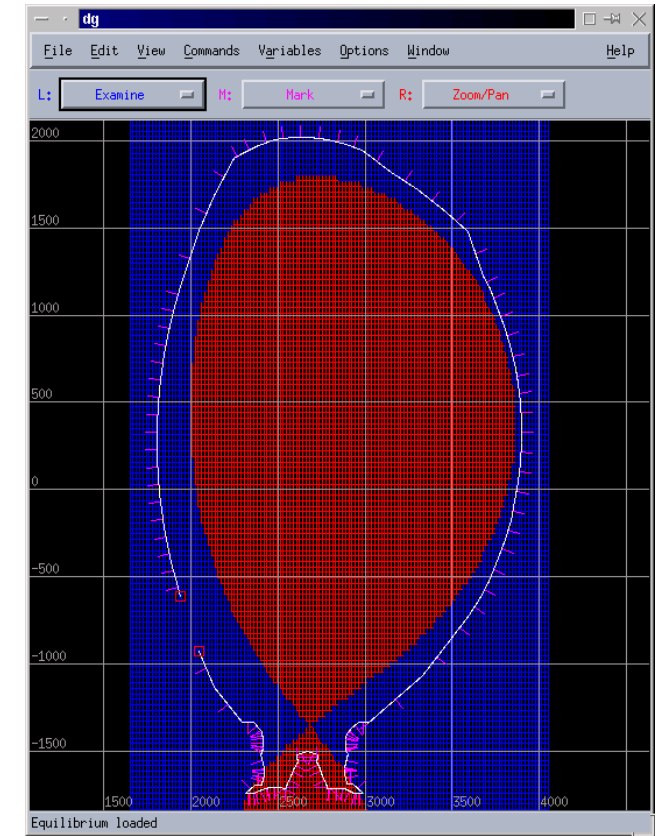
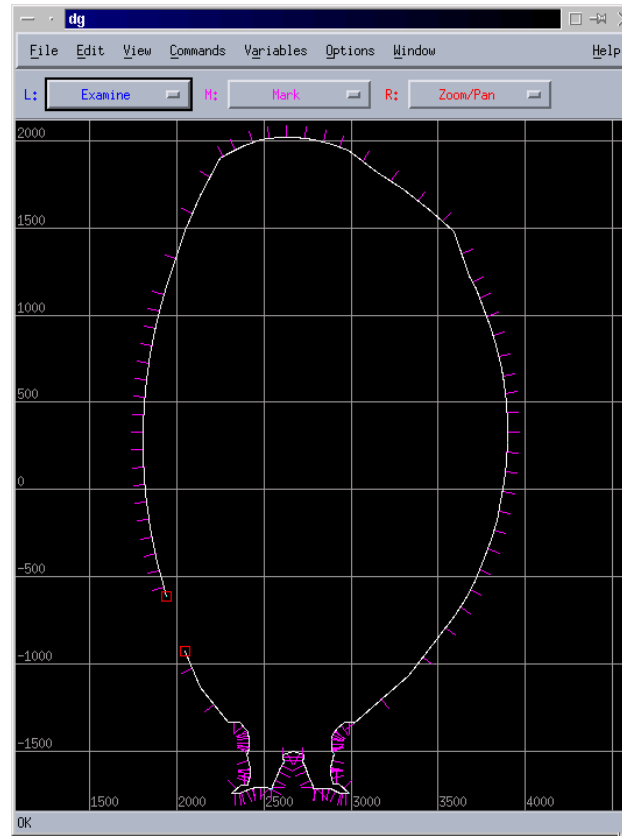
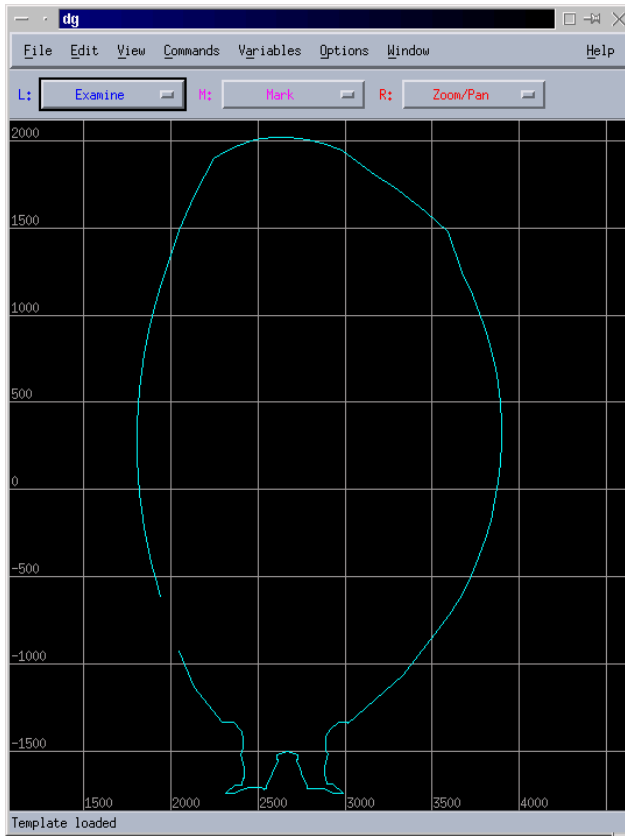
- a graphical tool for
 - developing new machines
 - for modifying present machines
 - for preparing the input files for
 - * Eirene
 - * B2 (SOLPS4.0)
 - * b2plot (mesh.extra)
- written by Andrei and Konstantin Kukushkin

- powerful graphical tool

```
setenv DEVICE jet  
sdg  
./dg
```

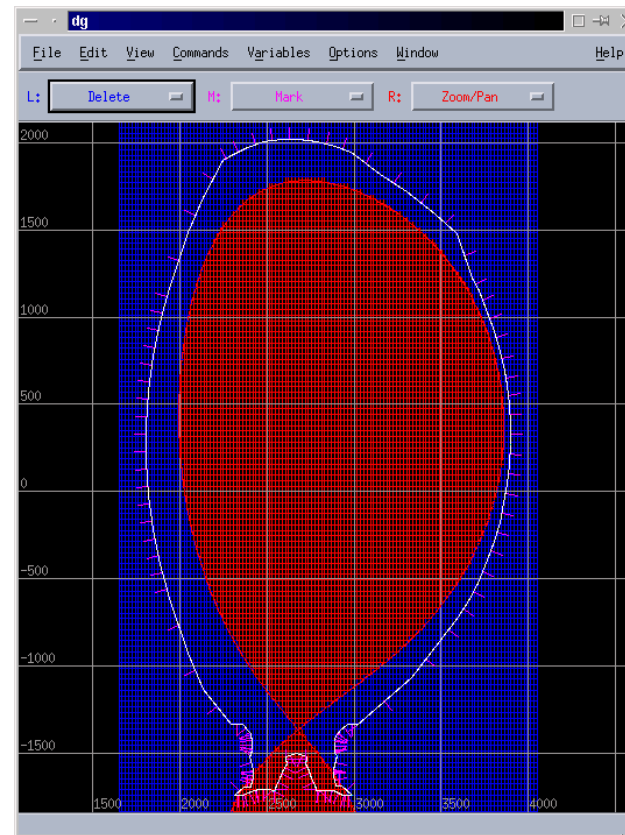
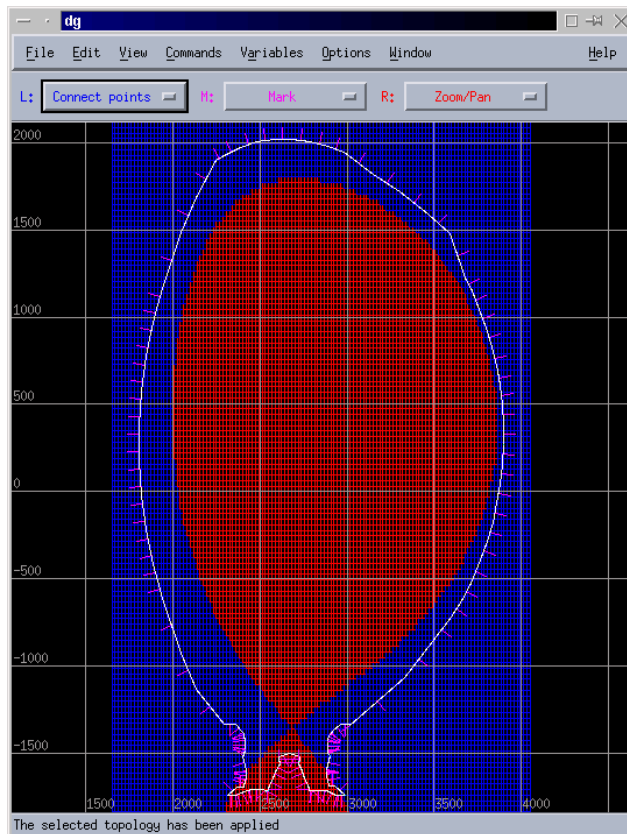


- File|Import|Template
- JET.IIGB.mm.ogr
- Commands|Convert|Template to elements
- File|Import|Equilibrium
- g045462.59381.txt.x4.equ



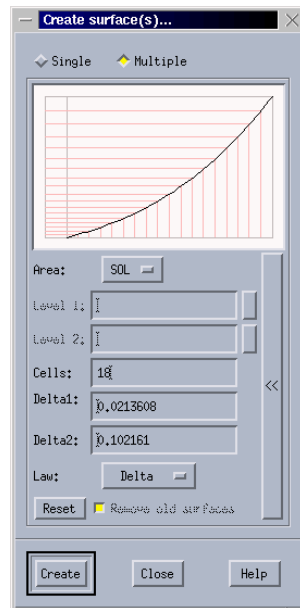
Joining up the vessel, setting topology & structure

- Change left button to “Connect points”
- Join up vacuum vessel
- File|Import|Topology
- SN
- Variable|Structure
- set “Structure”, “Inner Target” and “Outer Target”

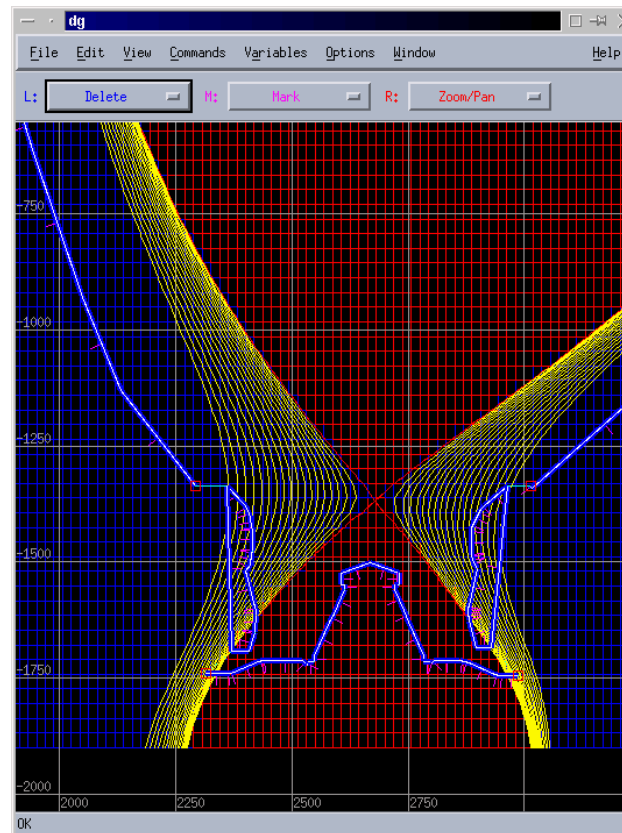


Creating flux surfaces

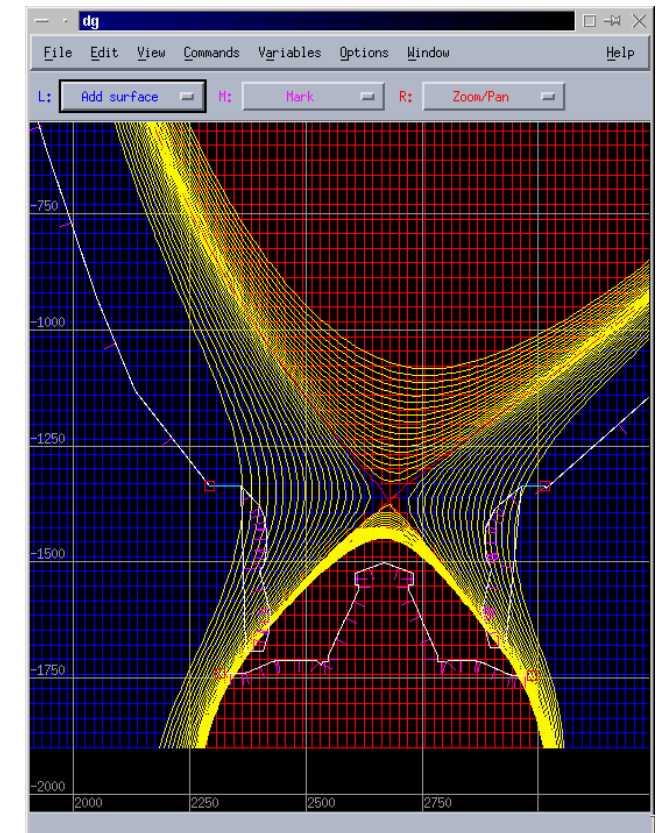
- Edit|Create|Surfaces
- set “Structure”, “Inner Target” and “Outer Target”



- Creating SOL surfaces

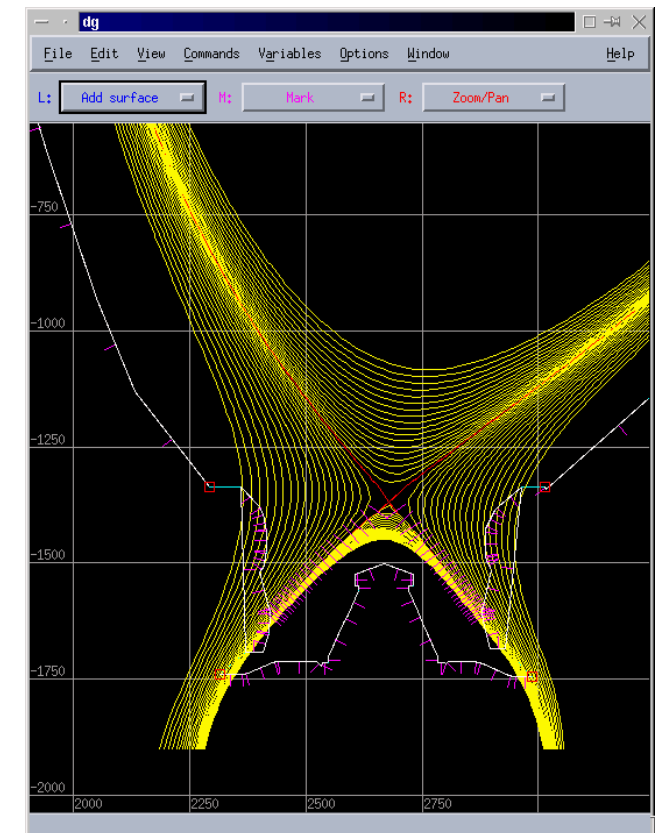
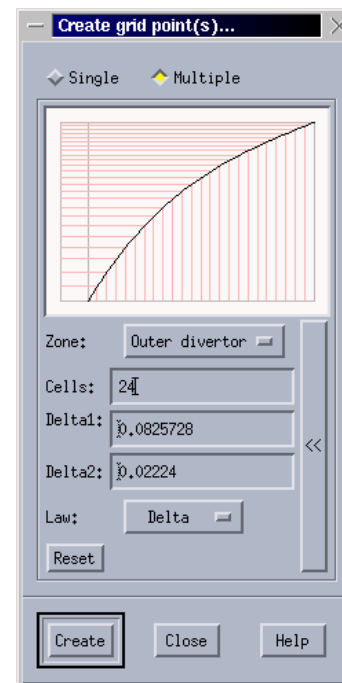
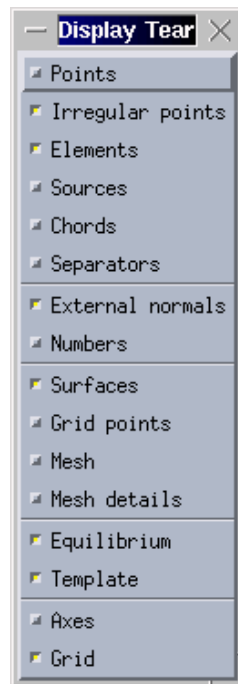


- Create Core and PFR surfaces

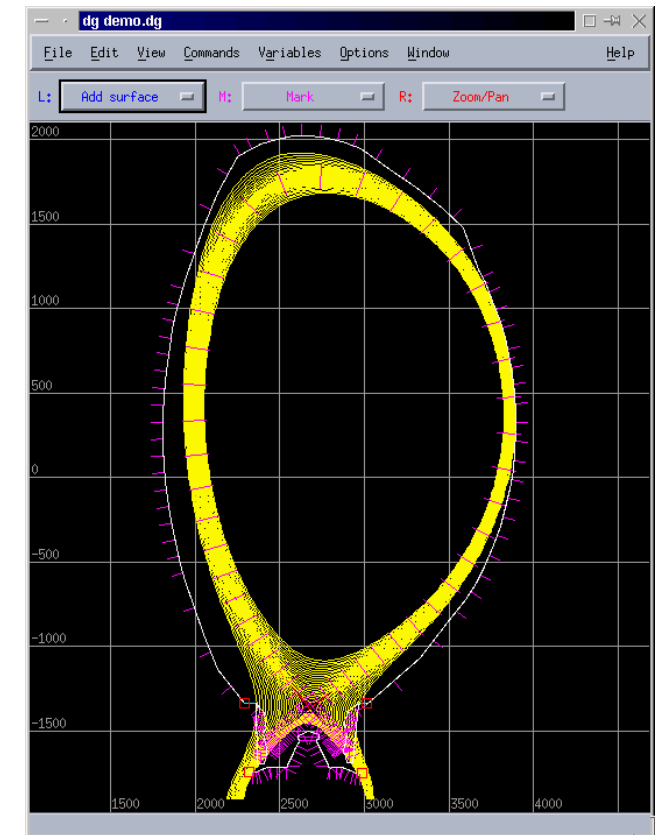
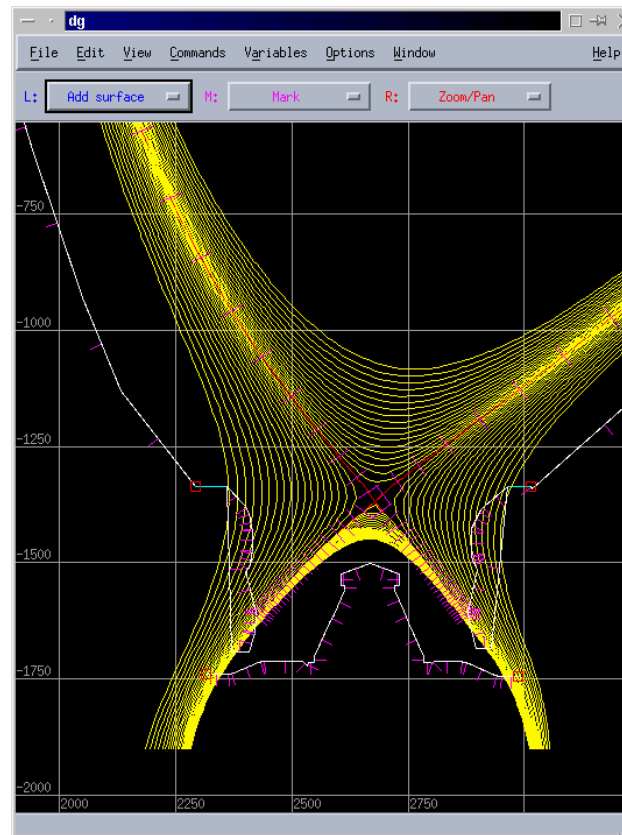
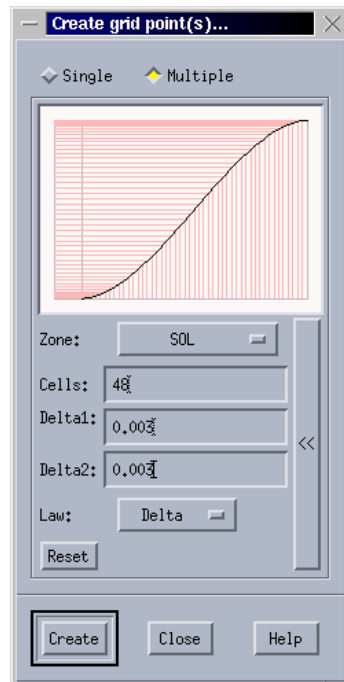


Creating orthogonals

- View|Display
- and tear off
- Edit|Create|Grid points
- Adjust so points bunch closer near target
- Result of creating inner and outer divertor grid points

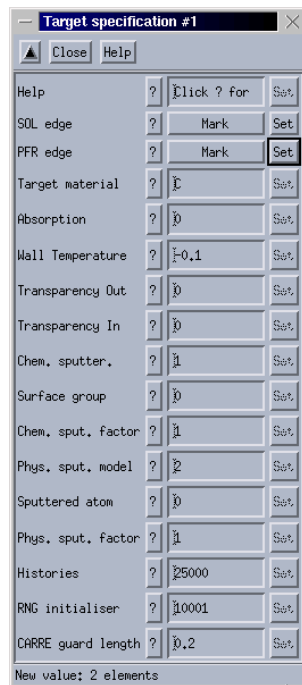


- Adjust so points bunch closer near separatrix
- Result
- Zoom out

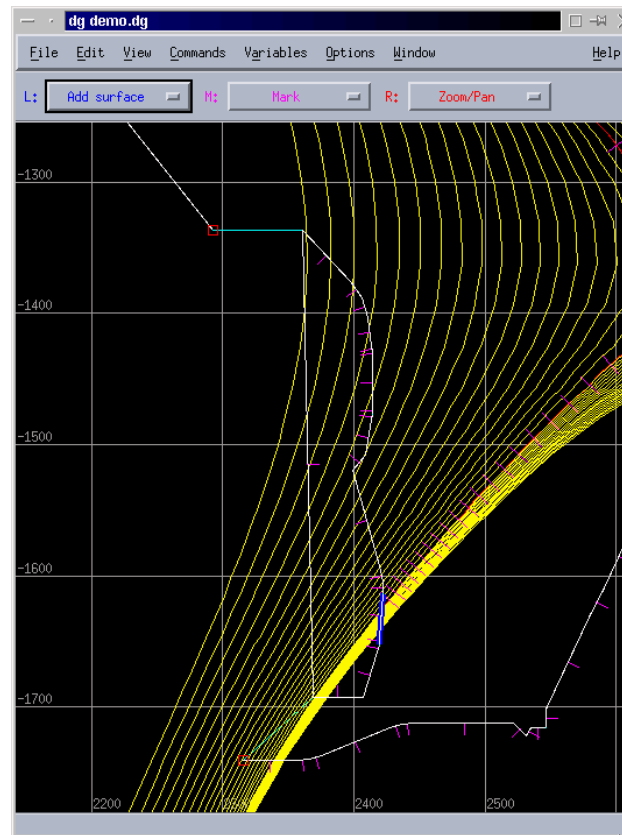


Setting targets, saving and outputting

- Variables|Target specification|#1



- Set SOL and PFR intersections

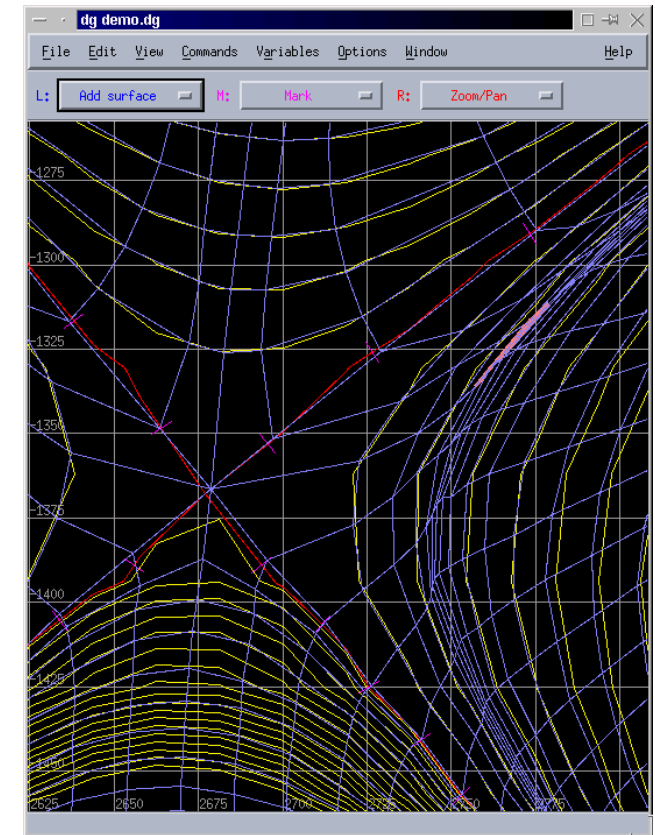
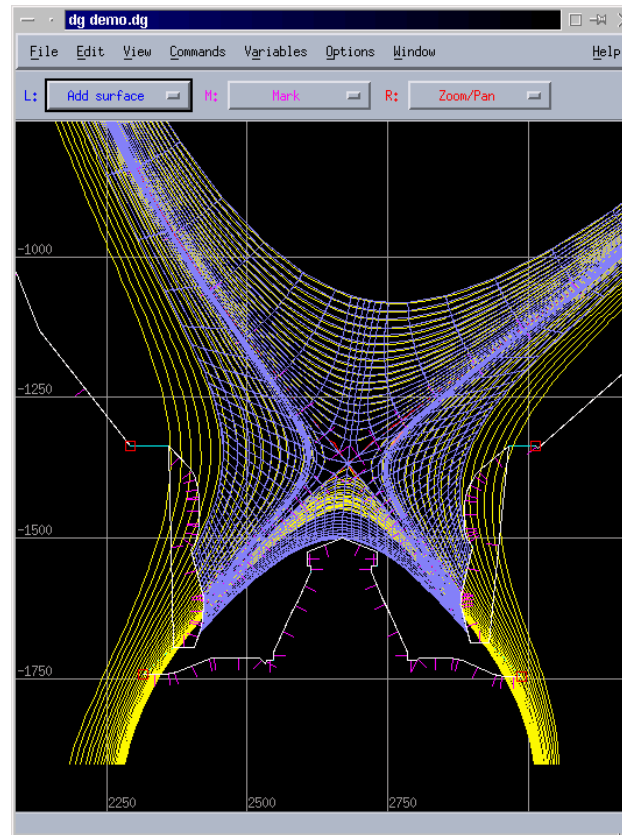
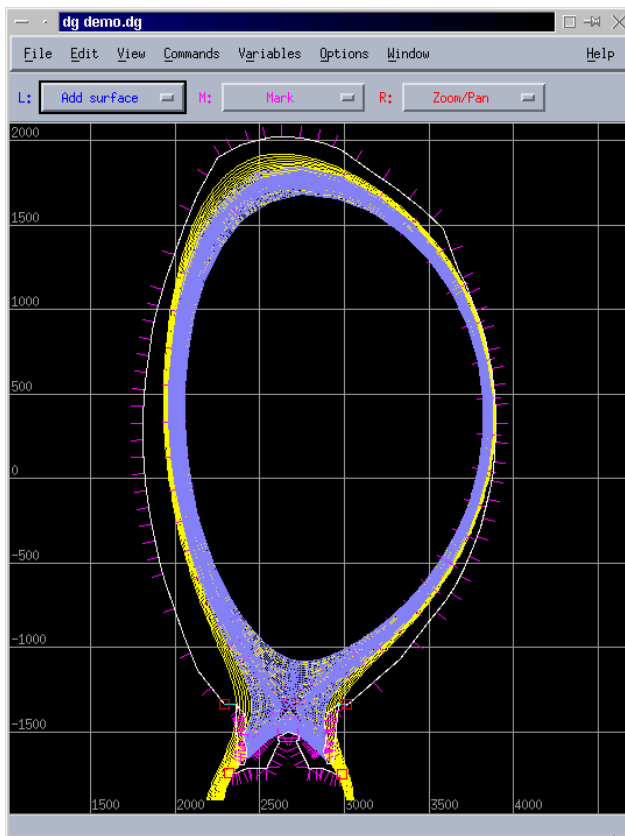


- File|Save
- In this case, as demo
- File|Output
- In the shell
 - ./Ins demo
 - ssc
 - ./carre -
- Type return at each question except when asked "Do you wish to accept these values (y/n)?" to which answer "y"
- In this case, got a message that the file was saved as "/space/dpc/afs/solps/solps4.0/src/Sonnet/output/ascii/jet/jet.carre.014"

- File|Import|Mesh
- jet.carre.014

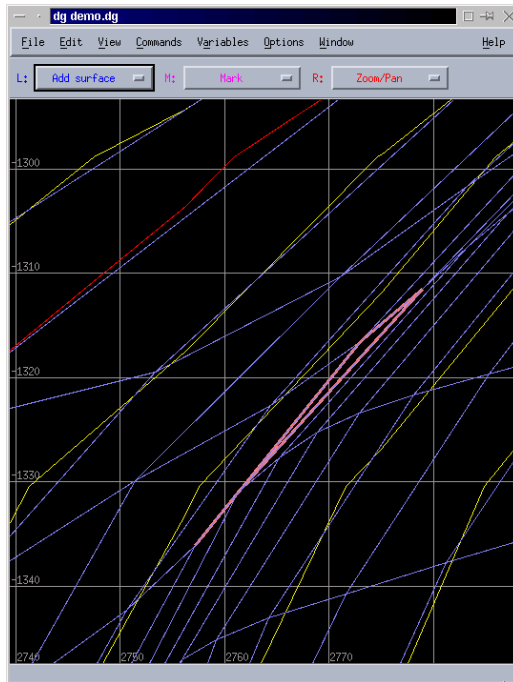
- Zoom in

- Zoom in again

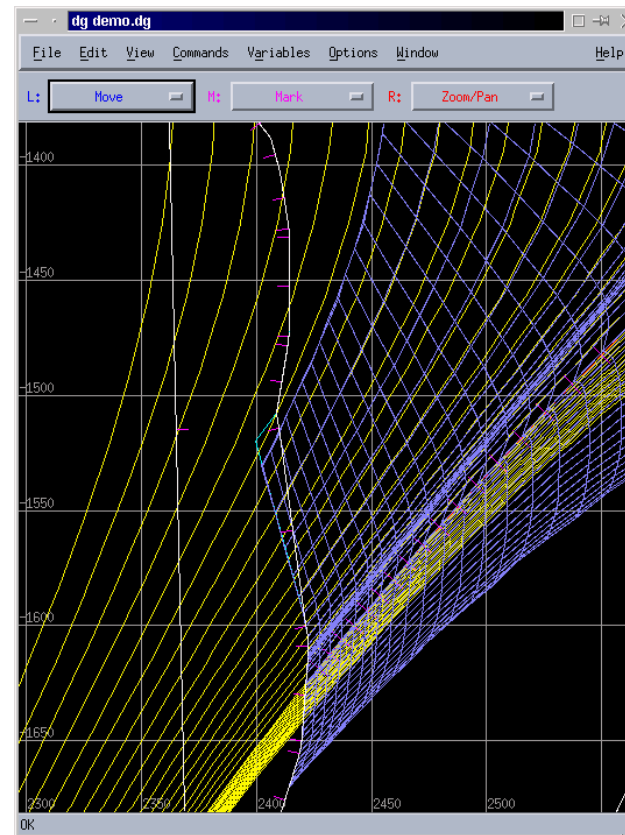


Modifying the targets

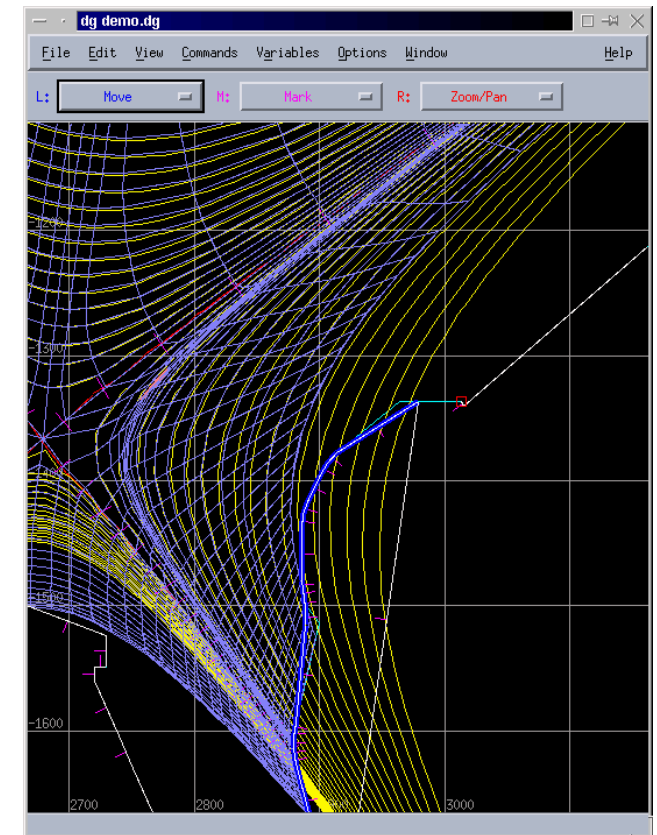
- Zoom again



- first the inner

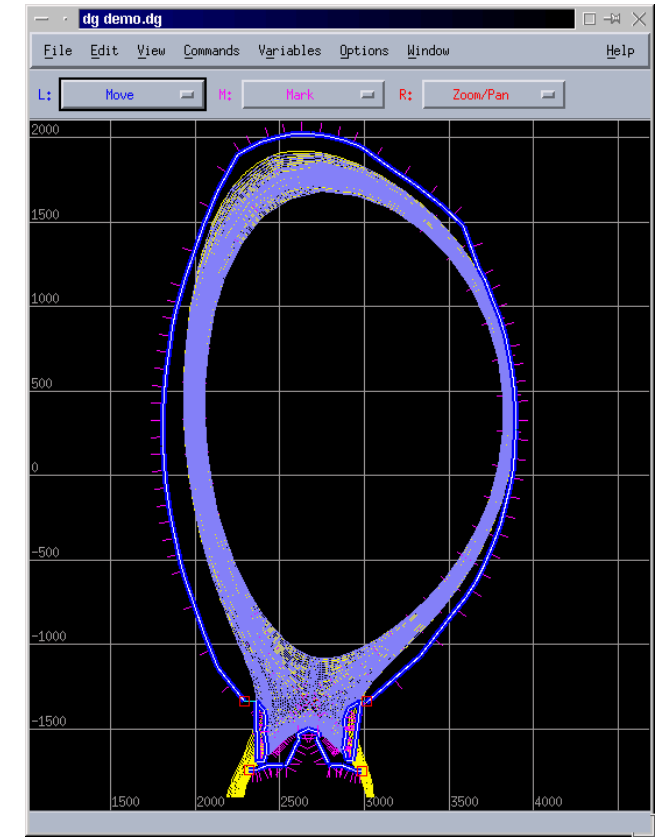
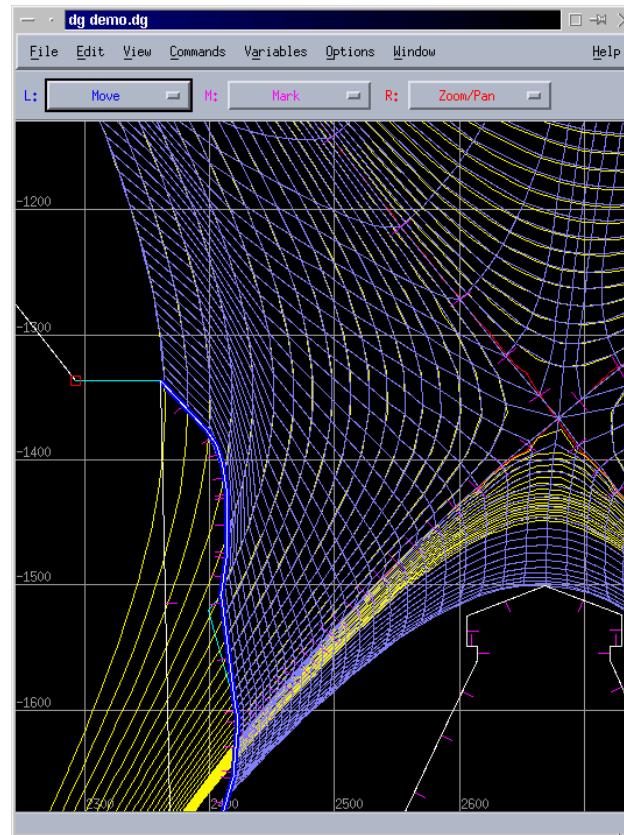
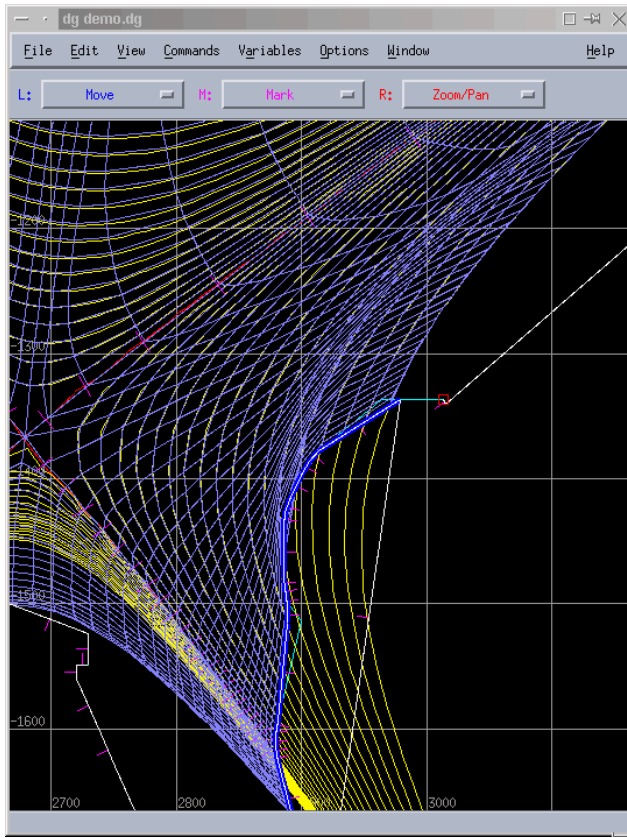


- and then the outer



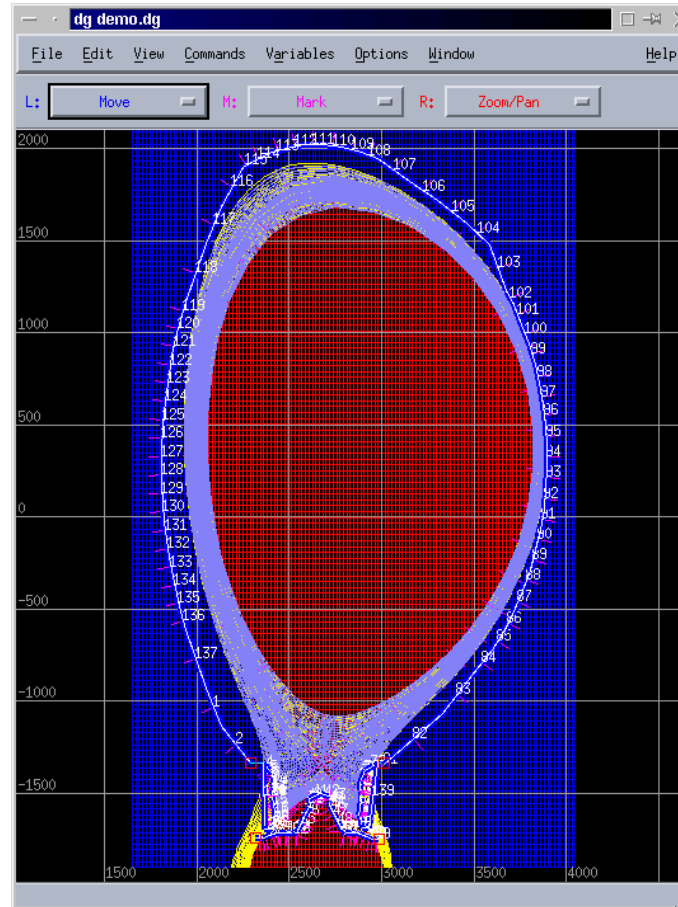
- The magenta piece indicates that there is a problem with the mesh
- so we can try modifying the targets

- redo the output, re-run Carre, import new mesh
- zoom out
- grid looks better



We have a grid!

- Switch on display of
 - numbers
 - equilibrium
- we have the final result



- we now have a mesh we could use for running
- it is probably easier, though, to use
 - aggj
 - grid2d